FINANCIAL SERVICES · DATA ENGINEERING

Deduplication and Entity Resolution

Medallion architecture, built with PySpark

Bronze Silver Gold

1 · TASK OVERVIEW

One customer, many records

Core banking

Smith, Bill

DOB 03/14/1985

(555) 014-2000

Credit card

WILLIAM SMITH

DOB 1985-03-14

5550142000

Loan

William Smith

DOB 14-Mar-1985

+1 555.014.2000

Deduplication

Same source, repeated record

Entity resolution

Different sources, same person

1 · TASK OVERVIEW

Why the bank cares

Missed sanctions hits

Split identities slip through screening

Inflated customer counts

Duplicates overstate reports

Wrong exposure

One customer’s products are split apart

Repeat KYC work

Existing customers onboarded again

2 · REQUIREMENTS

What the solution must do

One record per person

Trusted and complete

Raw data untouched

Bronze never changes

Every merge explainable

Rule and ID are stored

Accuracy measured

Precision and recall

3 · APPROACH

Medallion: three tables, three IDs

BRONZE

Raw, never changed

bronze_id · key

2,091 rows

SILVER

Clean and linked

silver_id · key

bronze_id · link back

entity_id · which person

2,091 rows, none deleted

GOLD

Best record per person

gold_id · key

entity_id · the person

silver_ids, bronze_ids · trace

1,100 rows, duplicates collapsed

Why layers: re-run anytime, explain every merge, trace to source

4 · IMPLEMENTATION

The pipeline: six steps

Step 1

Generate data

2,091 messy records, 3 sources

Step 2

Bronze

Load raw, change nothing

Step 3

Silver: standardize

One format per column

Step 4

Silver: match

Link records to an entity_id

Step 5

Gold

Best record per person

Step 6

Evaluate

Precision and recall

4 · IMPLEMENTATION

Silver step 1: standardize

Field	Raw (bronze)	Standard (silver)
Phone	(555) 014-2000 or +1 555.014.2000	5550142000
Date of birth	03/14/1985 or 14-Mar-1985	1985-03-14
Name	Smith, Bill	WILLIAM SMITH
State	Texas	TX
Street	12 Main Street	12 MAIN ST
Email	BILL@MAIL.COM	bill@mail.com

Can’t be read? It becomes blank, never a guess.

4 · IMPLEMENTATION

Silver step 2: match

Step	What it does
Block	Compare only records with the same date of birth
Rule 1	Same SSN
Rule 2	Same phone and last name
Rule 3	Same email
Rule 4	Close first name, same last name and ZIP
Cluster	A matches B, B matches C: one person
Assign	Every group gets an entity_id

Blank value = no evidence, never a mismatch.

4 · IMPLEMENTATION

Gold: choose the best value

Field	Rule
Name, DOB, SSN, phone, email	Most frequent value, ignoring blanks
Address	From the most recent record
Counts	Records merged, systems involved
Trace	silver_ids and bronze_ids listed

Gold never invents data.

5 · CHALLENGES

What got in the way

Setup

- PySpark needs Java 17
- Python 3.14 too new, used 3.12
- Same column name broke a join
- Wrong folder paths broke steps

Matching

- Too many pairs: block by birth date
- Missing values: no evidence
- Typos survive cleaning: fuzzy match
- Chains can merge strangers

Wrong merge or missed match?

Screening favors recall. Merging favors precision.

6 · TESTING

Test against the known answer

Synthetic data has an answer key: the true person for every record. We judge every pair of records.

Right merge

Grouped together.
Same person.

Wrong merge

Grouped together.
Different people.

Missed match

Kept apart.
Same person.

Precision: of the merges made, how many were right?

Recall: of the real duplicates, how many did we find?

6 · TESTING

Results

1,100 Entities found	**1,100** Real people (answer key)	**2,091** Source records in
1.00 Precision	**1.00** Recall	**0 / 0** Wrong merges / missed matches

A perfect score on mild noise. Next test: harder data.

7 · CURRENT STATUS

Where we are

Item	Status
Use case and design	Done
Environment: Java, Python, PySpark	Done
Synthetic data, 2,091 records	Done
Bronze and silver (standardize, match)	Done
Gold, 1,100 people	Done
Evaluation: precision 1.00, recall 1.00	Done
Postgres tables, CSV exports, SQL practice	Done
Harder, noisier test data	Next

8 · KEY LEARNINGS

What I learned

Clean first

Formats break matching

Never delete in silver

Link by ID, collapse in gold

Blank is not a mismatch

Missing data adds no evidence

Block to scale

Compare only likely pairs

Precision vs recall

A business choice

Ground truth

Scores rules, but needs harder data

9 · NEXT STEPS

What comes next

Now

- Add harder noise to the data
- Re-test precision and recall
- Tune the rules

Then

- Scored matching with review queue
- Household linking
- Match new records daily

Later

- Companies and beneficial owners
- Sanctions watchlist screening
- Scale up on a cluster